📝 Title

Retail Business Performance & Profitability Analysis

📌 1. Objective

The objective of this project is to analyze retail transaction data to identify profit trends, loss-making products, and optimize business decisions using data-driven insights.

🛠 2. Tools Used

- SQL – Data cleaning and analysis
- Python (Pandas, Matplotlib, Seaborn) – Data processing and correlation analysis
- Tableau – Data visualization and dashboard creation

📊 3. Data Description

The dataset contains retail sales data including:

- Category and Sub-category
- Sales and Profit
- Order Date
- Region
- Discount

🔍 4. Analysis Performed

SQL Analysis:

- Calculated profit by category
- Identified loss-making sub-categories
- Analyzed regional performance

Python Analysis:

- Created correlation between inventory days and profit
- Visualized relationship using scatter plot

Tableau Dashboard:

- Built interactive dashboard
- Included profit analysis, loss products, and sales trends

📈 5. Key Insights

- Technology category generates highest profit
- Furniture category shows low profitability
- Tables sub-category results in losses
- High discounts reduce profit margins
- Inventory days have very weak correlation with profit

💡 6. Recommendations

- Reduce discounts on loss-making products
- Focus on high-profit categories
- Improve pricing strategies
- Optimize inventory management

✅ 7. Conclusion

This project demonstrates how data analysis can help businesses improve profitability and make informed decisions using SQL, Python, and Tableau.